2

1	t_1	t_2	t_3	
s_1	1,1	2,0	3,3	x_1
s_2	0,3	3,2	4,1	x_2
s_3	2,2	4,3	1,4	$1-x_1-x_2$
	y_1	y_2	$1-y_1-y_2$	

Игрок 1.

$E_1(x_1)$: $y_1 + 2y_2 + 3 - 3y_1 - 3y_2$.

$= 3 - 2y_1 - y_2$.

$E_1(x_2)$: $3y_2 + 4 - 4y_1 - 4y_2$

$= 4 - 4y_1 - y_2$.

$E_1(x_3)$: $2y_1 + 4y_2 + 1 - y_1 - y_2$

$= 1 + y_1 + 3y_2$.

Игрок 2.

$E_2(t_1)$: $x_1 + 3x_2 + 2 - 2x_1 - 2x_2$.

$= 2 - x_1 + x_2$.

$E_2(t_2)$: $0x_1 + 2x_2 + 3 - 3x_1 - 3x_2$

$= 3 - 3x_1 - x_2$.

$E_2(t_3)$: $3x_1 + x_2 + 4 - 4x_1 - 4x_2$.

$E_2(t_3) = 4 - x_1 - 3x_2$.

~~For pla~~ player 1 attaches probability y_1 to x_1, x_2, x_3, when.

$E_1(x_1) = E_1(x_2) = E_1(x_3)$

$E_1(x_1) = E_1(x_2)$

$\Rightarrow 3 - 2y_1 - y_2 = 4 - 4y_1 - y_2$.

$\Rightarrow y_1 = 1/2$.

$\Rightarrow E_1(x_2) = E_1(x_3)$

$\Rightarrow 4 - 4y_1 - y_2 = 1 + y_1 + 3y_2$.

$\Rightarrow 3 = 5y_1 + 4y_2$.

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if $y_1 = 1/2$. $\Rightarrow 3 = \frac{5}{2} + 4y_2$.

$\Rightarrow \frac{1}{2} = 4y_2$.

$\Rightarrow \frac{1}{8} = y_2$

when $y_1 = 1/2$, $y_2 = 1/8$, $1 - y_1 - y_2 = 1 - \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$.

$E_1(x_1) = E_1(x_2) = E_1(x_3)$

Next: player 2 attaches probabilities to t_1, t_2, t_3 when:

$$E_2(t_1) = E_2(t_2) = E_2(t_3)$$

$$E_2(t_1) = E_2(t_2)$$

$$\Rightarrow \quad 2 - x_1 + x_2 = 3 - 3x_1 - x_2.$$

$$\Rightarrow \quad \underline{2x_1 + 2x_2 = 1.} \qquad -(II)$$

$$E_2(t_2) = E_2(t_3)$$

$$\Rightarrow \quad 3 - 3x_1 - x_2 = 4 - x_1 - 3x_2.$$

$$\Rightarrow \quad 2x_2 - 2x_1 = 1. \qquad -(III)$$

$$\Rightarrow \quad x_2 = \underline{1/2}. \quad \text{from (II) \& (III)}$$

$$\Rightarrow \quad x_1 = 0.$$

$$E_1(t_1) = E_2(t_3)$$

$$\Rightarrow \quad 2 - x_1 + x_2 = 4 - x_1 - 3x_2.$$

$$\Rightarrow \quad 4x_2 = 2$$

$$\Rightarrow \quad x_2 = 1/2.$$

$$\Rightarrow$$

$$x_1 = 0, \; x_2 = 1/2, \; 1 - x_1 - x_2 = 1/2$$

$$E_2(t_1) = \frac{5}{2} = E_2(t_2) = E_2(t_3)$$

When player 1 attaches $x_1 = 0, x_2 = 1/2, 1 - x_1 - x_2 = 1/2$ over t_1, t_2, t_3 then

Player 2 attacks probability from t_1, t_2, t_3.

when Player 2 ~~has~~ attacks $y_1 = 1/2, y_2 = 1/8, 1-y_1-y_2 = 3/8$

Player 2 attacks position probability over

Q. such that $x_1 = 0, x_2 = 1/2, 1-x_1-x_2 = 1/2$.

Mixed strategy each eqn.

$$(x_1, x_2, 1-x_1-x_2) = (0, 1/2, 1/2)$$

$$(y_1, y_2, 1-y_1-y_2) = (1/2, 1/8, 3/8)$$